

Liang, Introduction to Java Programming, Eighth Edition, (c) 2011 Pearson Education, Inc. All rights reserved. 0132130807

Motivasyon

, bir Java

, ve

. Bu

, pratik problemlerin programlama

. Bu problemlerle; Java ilkel (primitive) veri tiplerini, ini, sabitlerini, veri tiplerini, ini, ifadelerini, girdi ve ile ilgili

.



- (2.2).
- Scanner (2.3).
- identifiers) kullanmak (2.4).
- (2.5-2.6).
- Atama (Assignment) deyimi (statement) ve ifadeleri (expression) ile programlamak (2.6).
- (2.7).
- : byte, short, int, long, float, double, and char (2.8.1).
- java (2.8.2 2.8.3).
- (2.9).
- Short hand (2.10).
- (2.11).
- (2.12).
- Char (2.13).
- (2.14).
- Bir dizgeyi (String String (2.15).
- (2.16).
- syntax runtime logic debug (2.17).
- (GUI) JOptionPane input dialog boxes etmek (2.18).



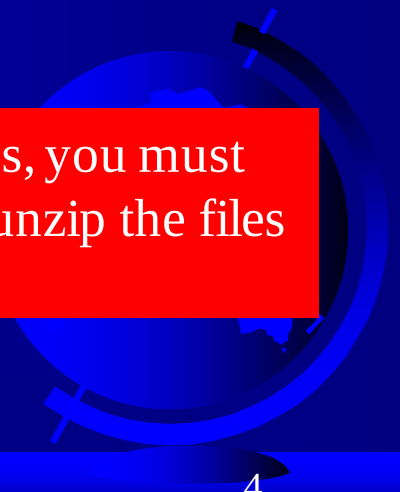
2.1

hesaplar

ComputeArea

Run

IMPORTANT NOTE: To enable the buttons, you must download the entire slide file *slide.zip* and unzip the files into a directory (e.g., c:\slide) .



```
public class ComputeArea {
  /** Main method */
  public static void main(String[] args) {
    double radius;
    double area;

    // Assign a radius
    radius = 20;

    // Compute area
    area = radius * radius * 3.14159;

    // Display results
    System.out.println("The area for the circle of radius " +
      radius + " is " + area);
  }
}
```

radius

allocate)

radius | no value



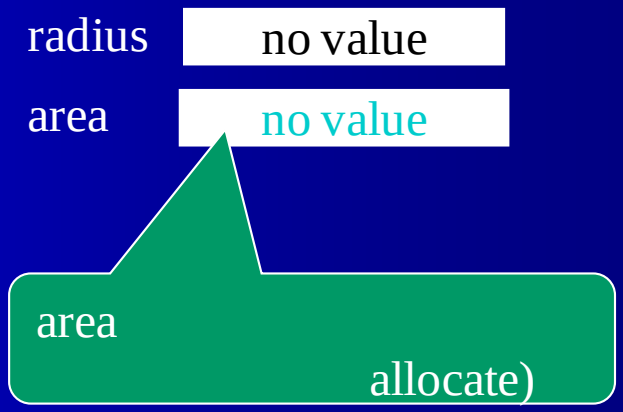
```
public class ComputeArea {
  /** Main method */
  public static void main(String[] args) {
    double radius;
    double area;

    // Assign a radius
    radius = 20;

    // Compute area
    area = radius * radius * 3.14159;

    // Display results
    System.out.println("The area for the circle of radius " +
      radius + " is " + area);
  }
}
```

Bellek (memory)



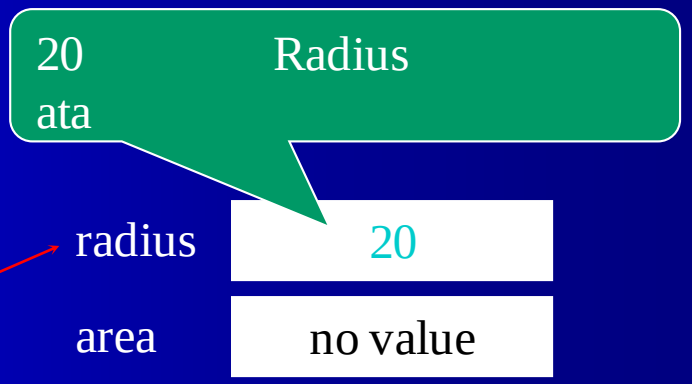


```
public class ComputeArea {
  /** Main method */
  public static void main(String[] args) {
    double radius;
    double area;

    // Assign a radius
    radius = 20;

    // Compute area
    area = radius * radius * 3.14159;

    // Display results
    System.out.println("The area for the circle of radius " +
      radius + " is " + area);
  }
}
```



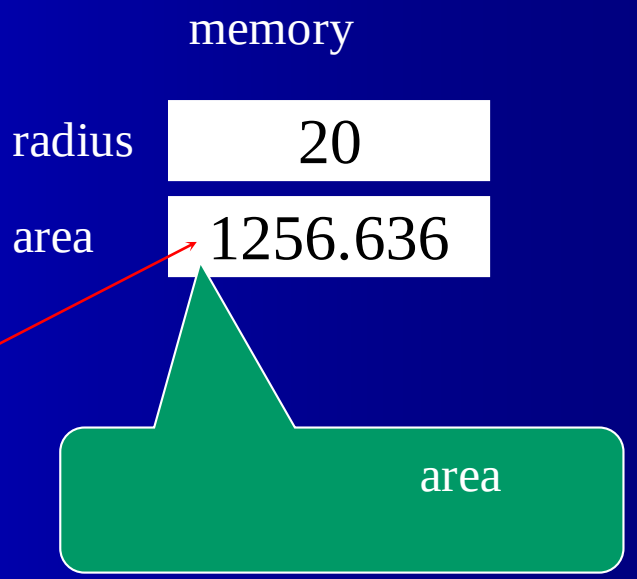


```
public class ComputeArea {
  /** Main method */
  public static void main(String[] args) {
    double radius;
    double area;

    // Assign a radius
    radius = 20;

    // Compute area
    area = radius * radius * 3.14159;

    // Display results
    System.out.println("The area for the circle of radius " +
      radius + " is " + area);
  }
}
```





```
public class ComputeArea {
  /** Main method */
  public static void main(String[] args) {
    double radius;
    double area;

    // Assign a radius
    radius = 20;

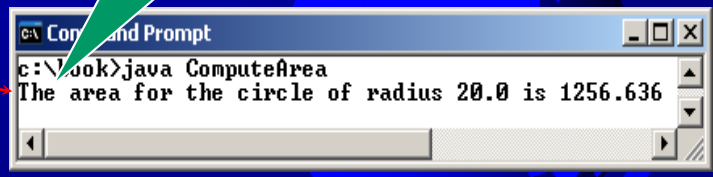
    // Compute area
    area = radius * radius * 3.14159;

    // Display results
    System.out.println("The area for the circle of radius " +
      radius + " is " + area);
  }
}
```

memory

radius	20
area	1256.636

Konsolda (Console)



Konsoldan (Console) Girdi (Input

1. Scanner nesnesi (object

```
Scanner input = new Scanner(System.in);
```

2. next(), nextByte(), nextShort(), nextInt(), nextLong(), nextFloat(), nextDouble(), or nextBoolean() to obtain to a string, byte, short, int, long, float, double, or boolean value. ,

```
System.out.print("Bir double değer giriniz:");
Scanner input = new Scanner(System.in);
double d = input.nextDouble();
```

ComputeAreaWithConsoleInput

Run

ComputeAverage

Run



Identifiers)

- harf, rakam, alt (_) ve dolar ($) bir karakter dizisidir.
- Bir , bir harf, alt (_) veya dolar ($) ile . Rakamla .
- Bir , bir kelime olamaz. (kelimelerin bir listesi Ek A, Java Anahtar Kelimeleri ne).
- Bir `true`, `false`, or `null` olamaz.
- Bir herhangi bir uzunlukta olabilir.



Variables)

```
// Compute the first area
radius = 1.0;
area = radius * radius * 3.14159;
System.out.println("The area is " +
  area + " for radius "+radius);

// Compute the second area
radius = 2.0;
area = radius * radius * 3.14159;
System.out.println("The area is " +
  area + " for radius "+radius);
```



(Declaring Variables)

```
int x;              // x' in bir tamsayı
                    // değişkeni olduğunu
                    // bildirme;
double radius;      // radius' un bir double
                    // değişkeni olduğunu
                    // bildirme;
char a;             // a' nın bir char
                    // değişkeni olduğunu
                    // bildirme;
```



(Assignment Statements)

```
x = 1;                  // 1 değerini X' e ata;

radius = 1.0;           // 1.0 değerini radius
                        // değişkenine ata;


a = 'A';                // 'A' değerini a
                        //değişkenine ata;
```



Bildirme ve Atama (Declaring and Initializing

- `int x = 1;`
- `double d = 1.4;`



Sabitler (Constants)

```
final datatype CONSTANTNAME = VALUE;

final double PI = 3.14159;
final int SIZE = 3;
```



(Numerical Data Types)

Name	Range	Storage Size
byte	-2^7 (-128) to 2^7-1 (127)	8-bit signed
short	-2^{15} (-32768) to $2^{15}-1$ (32767)	16-bit signed
int	-2^{31} (-2147483648) to $2^{31}-1$ (2147483647)	32-bit signed
long	-2^{63} to $2^{63}-1$ (i.e., -9223372036854775808 to 9223372036854775807)	64-bit signed
float	Negative range: -3.4028235E+38 to -1.4E-45 Positive range: 1.4E-45 to 3.4028235E+38	32-bit IEEE 754
double	Negative range: -1.7976931348623157E+308 to -4.9E-324 Positive range: 4.9E-324 to 1.7976931348623157E+308	64-bit IEEE 754

Numeric Operators)

Name	Meaning	Example	Result
+	Addition	34 + 1	35
-	Subtraction	34.0 - 0.1	33.9
*	Multiplication	300 * 30	9000
/	Division	1.0 / 2.0	0.5
%	Remainder	20 % 3	2

Integer Division)

+, -, *, /, ve %

5 / 2 → Bir (2) verir.

5.0 / 2 → Bir (2,5) sonucu verir.

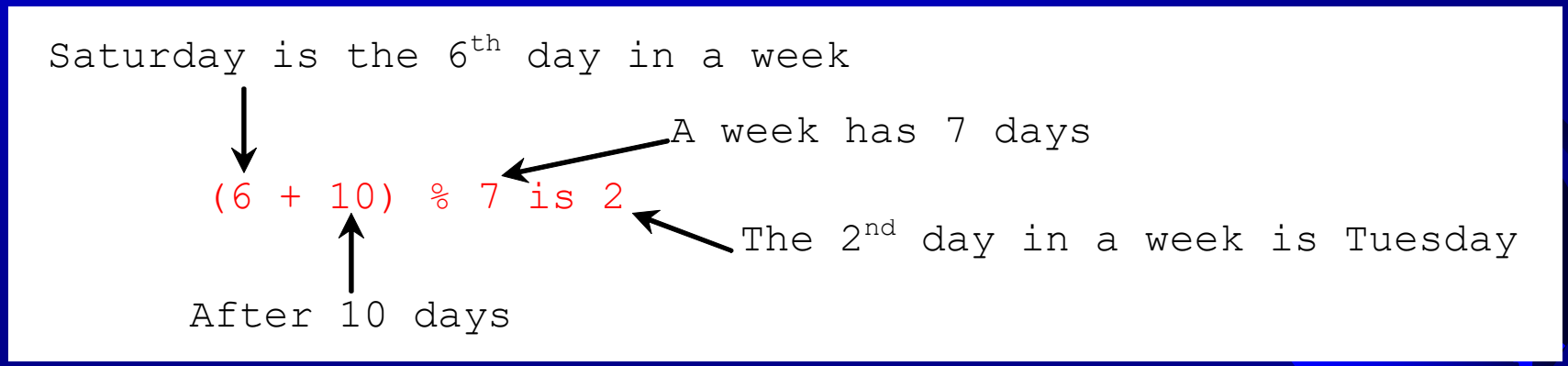
5 % 2 → 1 sonucunu verir.

sonucunda verir).



Remainder Operator)

Kalan programlamada . ,% 2 lar her zaman 0, tek lar ise her zaman 1 sonucunu verir. , bir mi yoksa tek mi belirlemek bu kullanabilirsiniz. cumartesi ve 10 . 10 hangi ? ifadeyi kullanarak o :



Problem:

DisplayTime

Run



NOT

Kayan (Floating-point) hesaplamalar, bu tam bir kaydedile . ,

System.out.println(1.0 - 0.1 - 0.1 - 0.1 - 0.1 - 0.1);

0.5 0.5000000000000001 .

System.out.println(1.0 - 0.9);

0.1 0.09999999999999998 . T tam olarak . Bu nedenle, hesaplamalar kesin bir sonucu verir.



Number Literals)

Literal

34, 1000000 ve

5.0

```
int i = 34;
long x = 1000000;
double d = 5.0;
```



Floating-Point Literals)

Kayan nokta basamakla . olarak, kayan nokta double tipinde olarak kabul edilir. 5.0 bir kayan (float) double olarak kabul edilir. F veya f harfini ekleyerek bir float yapabilir, d veya D harfini de ekleyerek bir double yapabilirsiniz. bir kayan 100.2f veya 100.2F ifadelerini, bir double ise 100.2d veya 100.2 ifadelerini kullanabilirsiniz.

Scientific Notation)

Kayan nokta bilimsel de belirtilebilir, 1.23456e2 ile olan 1.23456e + 2, 123.456'ya ve 1.23456e-2, 0.0123456'ya . E (veya e) bir temsil eder ve veya harf olabilir.



Arithmetic Expressions)

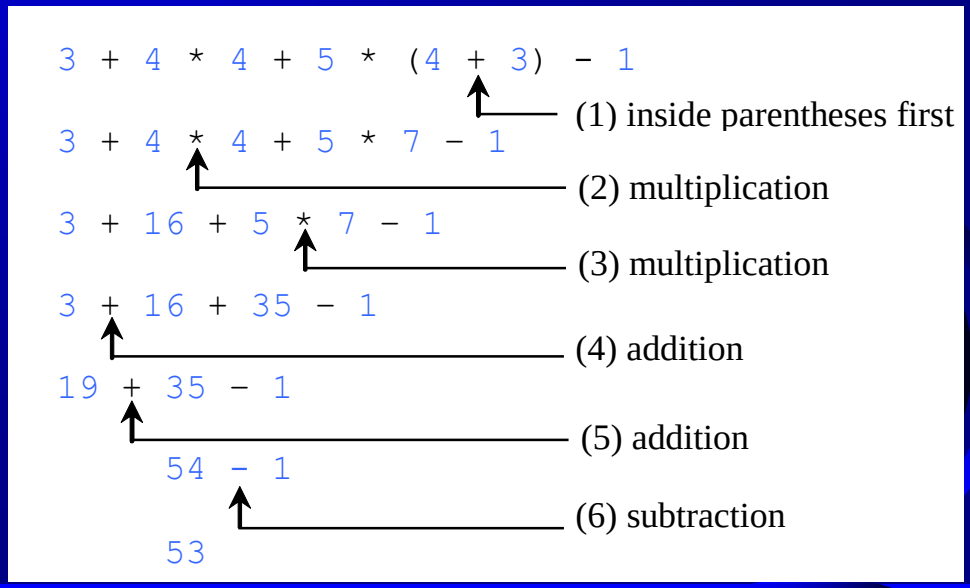
$$\frac{3+4x}{5} - \frac{10(y-5)(a+b+c)}{x} + 9(\frac{4}{x} + \frac{9+x}{y})$$

(3+4*x)/5 10*(y-5)*(a+b+c)/x + 9*(4/x + (9+x)/y)



Bir

Java, bir aritmetik ifadeyi kendi yoluna sahip olsa da, bir Java ifadesinin sonucu ve ona gelen aritmetik ifadesinin sonucu . Bu nedenle, bir Java ifadesini aritmetik uygulayabilirsiniz.



Problem:

Celsius kullanarak Fahrenheit program :

$$celsius = (\frac{5}{9})(fahrenheit - 32)$$

FahrenheitToCelsius

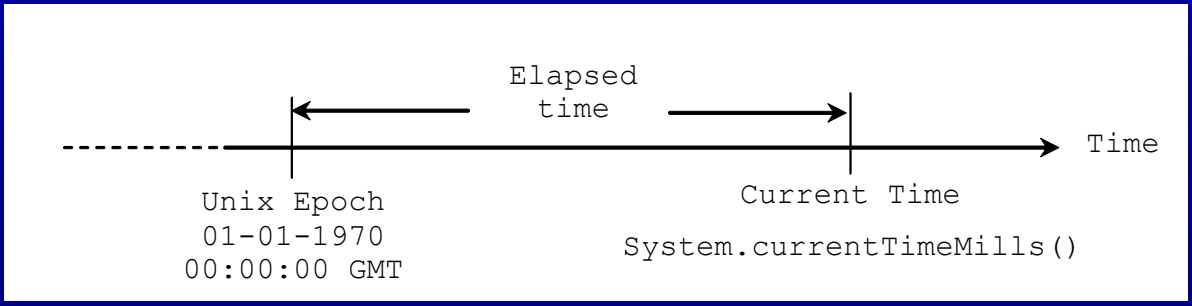
Run



Problem: Saati

GMT'de anki saat: dakika: saniye, yazan program 1:45:19 gibi).

CurrentTimeMill, System bir dir, 1 Ocak 1970 GMT gece bu yana milisaniye cinsinden verir. (1970, Unix sisteminin resmen saati elde etmek bu kullanabilir ve saniye, dakika ve saati gibi hesaplayabilirsiniz.



ShowCurrentTime

Run

Atama

İşleç	*Örnek*	*Eş değeri*
`+=`	`i += 8`	`i = i + 8`
`-=`	`f -= 8.0`	`f = f - 8.0`
`*=`	`i *= 8`	`i = i * 8`
`/=`	`i /= 8`	`i = i / 8`
`%=`	`i %= 8`	`i = i % 8`



	Ad	
++var	preincrement	The expression (++var) increments var by 1 and evaluates to the *new* value in var *after* the increment.
var++	postincrement	The expression (var++) evaluates to the *original* value in var and increments var by 1.
--var	predecrement	The expression (--var) decrements var by 1 and evaluates to the *new* value in var *after* the decrement.
var--	postdecrement	The expression (var--) evaluates to the *original* value in var and decrements var by 1.



```
int i = 10;
int newNum = 10 * i++;
```

Same effect as

```
int newNum = 10 * i;
i = i + 1;
```

```
int i = 10;
int newNum = 10 * (++i);
```

Same effect as

```
i = i + 1;
int newNum = 10 * i;
```

ve azaltma kullanmak ifadeleri , ancak bu ifadeleri ve zor bir hale getirebilir.

Bu , birden fazla ifadelerde veya bunun gibi birden kez kullanmaktan : int k = ++ i + i.



Atama ve Atama

Java 2'den , ifadeler statement olarak
Java 2'den bu yana,
ifadeler statement olarak :

variable op= expression; // Where op is +, -, *, /, or %

++variable;

variable++;

--variable;

variable--;



Numeric Type Conversion)

ifadeleri inceleyiniz:

```
byte i = 100;
long k = i * 3 + 4;
double d = i * 3.1 + k / 2;
```



(Conversion Rules)

tipte 2 operand binary (ikili) bir
Java kurallara operand
otomatik olarak .

1. Operand lardan biri double double

2. Operand lardan biri float float
3. Operand lardan biri long long
4. Aksi takdirde iki operand int

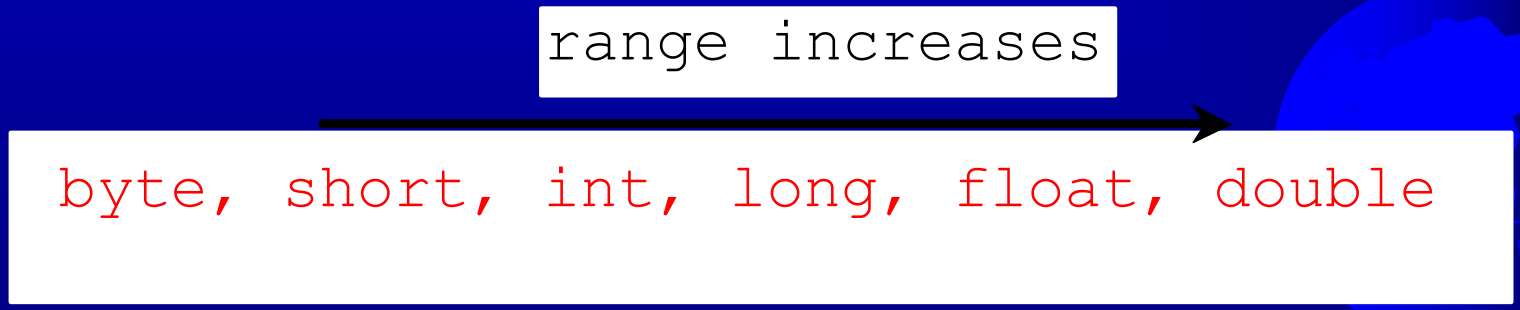


Tip (Type

```
Implicit casting
   double d = 3; (type widening)

Explicit casting
   int i = (int)3.0; (type narrowing)
   int i = (int)3.9; (Fraction part is
  truncated)
```

What is wrong? int x = 5 / 2.0;



Problem: Noktadan Sonra Rakam Tutmak

vergisini, sonra iki olarak
bir program .

SalesTax

Run



Problem: Bilgi Kredisi

Bu program, faiz , ve kredi girmesini ve ve toplam hesaplar.

$$monthlyPayment = \frac{loanAmount \times monthlyInterestRate}{1 - \frac{1}{(1 + monthlyInterestRate)^{numberOfYears \times 12}}}$$

ComputeLoan

Run

Karakter Veri Tipi (Character Data Type)

Four hexadecimal digits.

```
char letter = 'A'; (ASCII)
char numChar = '4'; (ASCII)
char letter = '\u0041'; (Unicode)
char numChar = '\u0034'; (Unicode)
```

NOT: ve azaltma bir sonraki veya bir Unicode karakterini elde etmek char de . ifadeler b karakterini .

```
char ch = 'a';
System.out.println(++ch);
```



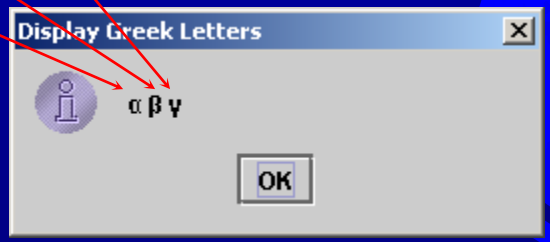
Unicode Format)

Unicode, '\ u0000' ile '\

\

Unicode \u03b1 \u03b2

\u03b3



Tanım	*Çıkış Tuşu*	*Unicode*
Backspace	`\b`	`\u0008`
Tab	`\t`	`\u0009`
Linefeed	`\n`	`\u000A`
Carriage return	`\r`	`\u000D`
Backslash	`\\`	`\u005C`
Single Quote	`\'`	`\u0027`
Double Quote	`\"`	`\u0022`



Ek B: ASCII Karakter

ASCII Karakter , Unicode'un bir alt ;

\ u0000 ile \ u007f

TABLE B.1 ASCII Character Set in the Decimal Index

	.0	.1	.2	.3	.4	.5	.6	.7	.8	.9	
0	nul	soh	stx	etx	eot	enq	ack	bel	bs	ht	0
1	nl	vt	ff	cr	so	si	dle	dc1	dc2	dc3	1
2	dc4	nak	syn	etb	can	em	sub	esc	fs	gs	2
3	rs	us	sp	!	"	#	$	%	&	'	3
4	(	)	*	+	,	-	.	/	0	1	4
5	2	3	4	5	6	7	8	9	:	;	5
6	<	=	>	?	@	A	B	C	D	E	6
7	F	G	H	I	J	K	L	M	N	O	7
8	P	Q	R	S	T	U	V	W	X	Y	8
9	Z	[	\	]	^	_	`	a	b	c	9
10	d	e	f	g	h	i	j	k	l	m	10
11	n	o	p	q	r	s	t	u	v	w	11
12	x	y	z	{	|	}	~	del			12

ASCII Karakter

ASCII Karakter , Unicode'un bir alt ;

\ u0000 ile \ u007f

TABLE B.2 ASCII Character Set in the Hexadecimal Index

	0	*1*	*2*	*3*	*4*	*5*	*6*	*7*	*8*	*9*	*A*	*B*	*C*	*D*	*E*	*F*
0	nul	soh	stx	etx	eot	enq	ack	bel	bs	ht	nl	vt	ff	cr	so	si
1	dle	dcl	dc2	dc3	dc4	nak	syn	etb	can	em	sub	esc	fs	gs	rs	us
2	sp	!	“	#	$	%	&	’	(	)	*	+	,	-	.	/
3	0	1	2	3	4	5	6	7	8	9	:	;	<	=	>	?
4	@	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
5	P	Q	R	S	T	U	V	W	X	Y	Z	[	\	]	^	_
6	‘	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o
7	p	q	r	s	t	u	v	w	x	y	z	{	|	}	~	del

Char ve Tipler Casting)

```
int i = 'a'; // Same as int i = (int)'a';

char c = 97; // Same as char c = (char)97;
```



Problem: Parasal Birimler

Bu program ve temsil eden girmesini ve parasal tek dolar, , dime (10 cent), nickel (5 cent) ve penny (1 cent) cinsinden listeleyen bir rapor .

, bu maksimum dolar , maksimum vb. Bildirmelidir.

ComputeChange

Run



Hesaplama

Amount 11.56 olarak kabul edin.

```
int remainingAmount = (int)(amount * 100);

// Find the number of one dollars
int numberOfOneDollars = remainingAmount / 100;
remainingAmount = remainingAmount % 100;

// Find the number of quarters in the remaining amount
int numberOfQuarters = remainingAmount / 25;
remainingAmount = remainingAmount % 25;

// Find the number of dimes in the remaining amount
int numberOfDimes = remainingAmount / 10;
remainingAmount = remainingAmount % 10;

// Find the number of nickels in the remaining amount
int numberOfNickels = remainingAmount / 5;
remainingAmount = remainingAmount % 5;

// Find the number of pennies in the remaining amount
int numberOfPennies = remainingAmount;
```

remainingAmount 1156

remainingAmount



Hesaplama

Amount 11.56 olarak kabul edin.

```
int remainingAmount = (int)(amount * 100);

// Find the number of one dollars
int numberOfOneDollars = remainingAmount / 100;
remainingAmount = remainingAmount % 100;

// Find the number of quarters in the remaining amount
int numberOfQuarters = remainingAmount / 25;
remainingAmount = remainingAmount % 25;

// Find the number of dimes in the remaining amount
int numberOfDimes = remainingAmount / 10;
remainingAmount = remainingAmount % 10;

// Find the number of nickels in the remaining amount
int numberOfNickels = remainingAmount / 5;
remainingAmount = remainingAmount % 5;

// Find the number of pennies in the remaining amount
int numberOfPennies = remainingAmount;
```

remainingAmount	1156
numberOfOneDollars	11

numberOfOneDollars



Hesaplama

Amount 11.56 olarak kabul edin.

```
int remainingAmount = (int)(amount * 100);

// Find the number of one dollars
int numberOfOneDollars = remainingAmount / 100;
remainingAmount = remainingAmount % 100;

// Find the number of quarters in the remaining amount
int numberOfQuarters = remainingAmount / 25;
remainingAmount = remainingAmount % 25;

// Find the number of dimes in the remaining amount
int numberOfDimes = remainingAmount / 10;
remainingAmount = remainingAmount % 10;

// Find the number of nickels in the remaining amount
int numberOfNickels = remainingAmount / 5;
remainingAmount = remainingAmount % 5;

// Find the number of pennies in the remaining amount
int numberOfPennies = remainingAmount;
```

remainingAmount 56

numberOfOneDollars 11

remainingAmount



Hesaplama

Amount 11.56 olarak kabul edin.

```
int remainingAmount = (int)(amount * 100);

// Find the number of one dollars
int numberOfOneDollars = remainingAmount / 100;
remainingAmount = remainingAmount % 100;

// Find the number of quarters in the remaining amount
int numberOfQuarters = remainingAmount / 25;
remainingAmount = remainingAmount % 25;

// Find the number of dimes in the remaining amount
int numberOfDimes = remainingAmount / 10;
remainingAmount = remainingAmount % 10;

// Find the number of nickels in the remaining amount
int numberOfNickels = remainingAmount / 5;
remainingAmount = remainingAmount % 5;

// Find the number of pennies in the remaining amount
int numberOfPennies = remainingAmount;
```

remainingAmount	56
numberOfOneDollars	11
numberOfOneQuarters	2

numberOfOneQuarters



Hesaplama

Amount 11.56 olarak kabul edin.

```
int remainingAmount = (int)(amount * 100);

// Find the number of one dollars
int numberOfOneDollars = remainingAmount / 100;
remainingAmount = remainingAmount % 100;

// Find the number of quarters in the remaining amount
int numberOfQuarters = remainingAmount / 25;
remainingAmount = remainingAmount % 25;

// Find the number of dimes in the remaining amount
int numberOfDimes = remainingAmount / 10;
remainingAmount = remainingAmount % 10;

// Find the number of nickels in the remaining amount
int numberOfNickels = remainingAmount / 5;
remainingAmount = remainingAmount % 5;

// Find the number of pennies in the remaining amount
int numberOfPennies = remainingAmount;
```

remainingAmount	6
numberOfOneDollars	11
numberOfQuarters	2

remainingAmount



Dizge (String) Tipi

Char bir karakteri temsil eder. Bir karakter dizesini temsil etmek , String veri . ,
String message = "Welcome to Java";

Dizge Java System ve JOptionPane gibi bir . Dizge ilkel bir . Referans olarak bilinir. Herhangi bir Java bir referans olarak . Referans veri tipleri, ve 7. olarak ele . bir String (declare), bir stringin ve dizgelerin bilmeniz yeterlidir.



String Concatenation)

```
// String
String message = "Welcome " + "to " + "Java";

// String Chapter
String s = "Chapter" + 2; // s becomes Chapter2

// String Supplement
String s1 = "Supplement" + 'B'; // s1 becomes SupplementB
```



Programlama Stili ve Belgeler

- Uygun Yorumlar
-
- Girintileme ve
- Blok Stilleri



Uygun Yorumlar

ne , temel ,
destekleyici veri ve benzersiz
teknikleri bir
ekleyin.

, ,
, tarihinizi ve bir
ekleyin.



- ve isimler .
- ve

harf . isim
kelimeden , hepsini bir arada
, ilk kelime harf
ve sonraki her kelimenin ilk harfini
harfle . , `radius` ve `area`
ve `computeArea` gibi.



- (Class) :

  smindeki her kelimenin ilk harfini
  .
  , `ComputeArea` gibi.

- Sabitler:

  `PI` ve
  MAX_VALUE sabitleri gibi.



Girintileme ve

- Girintili Yazma

.

- (Spacing)

Kodun

..



Blok Stilleri

Parantezler sonu (End-of-line) stilini

.

Next-line style

```
public class Test
{
  public static void main(String[] args)
  {
    System.out.println("Block Styles");
  }
}
```

End-of-line style

```
public class Test {
  public static void main(String[] args) {
    System.out.println("Block Styles");
  }
}
```

Programlama (Programming Errors)

- (Syntax)

Derleyici (Compiler) .

- (Runtime)

iptal edilmesine neden olurlar.

- (Logic)

ler.



Syntax

```
public class ShowSyntaxErrors {
  public static void main(String[] args) {
    i = 30;
    System.out.println(i + 4);
  }
}
```



Runtime

```
public class ShowRuntimeErrors {
  public static void main(String[] args) {
    int i = 1 / 0;
  }
}
```



Logic

```
public class ShowLogicErrors {
  // Determine if a number is between 1 and 100 inclusively
  public static void main(String[] args) {
    // Prompt the user to enter a number
    String input = JOptionPane.showInputDialog(null,
      "Please enter an integer:",
      "ShowLogicErrors", JOptionPane.QUESTION_MESSAGE);
    int number = Integer.parseInt(input);

    // Display the result
    System.out.println("The number is between 1 and 100, " +
      "inclusively? " + ((1 < number) && (number < 100)));

    System.exit(0);
  }
}
```

Debugging)

(bug) denilmektedir.

bulma ve ise hata (debugging)
verilir. Hata ortak bir
program daraltmak bir
kombinasyonunun . elle
izleyebilir (yani, okuyarak
yakalayabilirsiniz) veya veya
ifadeler
(print statement) ekleyebilirsiniz. Bu basit
bir program yarayabilir. Ancak
bir program hata en etkili bir
hata
.



Debugger)

Hata , hata bir
. Bir hata
kullanabilirsiniz;

- Bir .
-
-
-
-
-



Dizgeleri (Strings) Integers

kutusundan bir dizgedir. 123 gibi bir girerseniz, 123 . olarak almak bir dizgeyi gerekir.

Bir dizgeyi (String) int , Integer statik parseInt gibi kullanabilirsiniz:

int intValue = Integer.parseInt(intString);

where intString is a numeric string such as 123 .



Dizgeleri (Strings)

Bir dizgyi (String) double , Double statik parseDouble gibi kullanabilirsiniz:

```
double doubleValue =Double.parseDouble(doubleString);
```

where doubleString



Problem: Kullanarak Kredi Hesaplama

Kredi programdakiyle , ancak girdi girdi girilmesi ve kutusunda gerekmektedir.

$$\frac{loanAmount \times monthlyInterestRate}{1 - \frac{1}{(1 + monthlyInterestRate)^{numberOfYears \times 12}}}$$

ComputeLoanUsingInputDialog

Run